

Quantum Peak Challenge: Session Report

Local Codex run on `quantum-junction`

6 June 2026

Abstract

This report consolidates the fragmented analysis performed in this session: static QASM forensics, exact statevector baselines, Aer MPS sampling, MPS distillation across seeds and bond dimensions, and the `peaked-circuit-simulation` MPO/unswapping attempt. The central result is that exact simulation validated ten small circuits; approximate methods agreed on most exact-checkable cases but `24_13` exposed a real failure mode of single-qubit marginal thresholding. For larger circuits, the safest outputs so far are repeated-sample candidates from MPS sampling/distillation, not single marginal-threshold strings.

1 Challenge and Bit Order

The challenge circuits are peaked: for each QASM circuit, the task is to recover the output bitstring with the largest measurement probability from input state $|0 \cdots 0\rangle$. The instructions state that Qiskit-style count strings are ordered with the right-most bit corresponding to qubit q_0 . All bitstrings in this report use that convention.

The challenge construction described in `INSTRUCTIONS.md` hides an RX-encoded target pattern inside identity-like blocks $U_i U_i^\dagger$ and then applies an obfuscation pass. The most relevant downloaded references were the peaked-circuit construction paper, the Kremer–Dupuis classical MPO/unswapping attack, and a 2D tensor-network sampling paper. The practical implication for this repo was: exact statevector is reliable only for small n , MPS sampling can work when the peak remains strong, and the local `peaked-circuit-simulation` implementation is a plausible structural attack but needs careful validation.

2 Artifacts Produced

The project environment was installed in the repo and all heavy runs were submitted to Slurm rather than run directly on the login node. The main output locations are:

- `agent_work/exact_baseline/`: exact Aer statevector results for circuits up to 28 qubits.
- `agent_work/static_forensics/`: QASM metrics, pair repetitions, angle summaries, operation-sequence similarity.
- `agent_work/algebraic_simplify/`: local simplification and Qiskit transpiler probes.
- `agent_work/mps_distill/`: Aer MPS sampling pilot over several seeds, shot counts, and bond dimensions.
- `outputs/sim_11_26_34_41_49/`: requested statevector/MPS runs for challenge IDs 11, 26, 34, 41, and 49, with distribution plots.
- `outputs/peaked_circuit_sim_pilot/` and `outputs/peaked_circuit_sim_all/`: peaked-circuit-simulation pilot and partial full sweep.

3 Static and Algebraic Analysis

Static parsing covered all 49 challenge QASM files. The gate set was only `rx`, `rz`, `cx`, and sparse `swap`; no measurements were present in the original challenge files. Repeated undirected CNOT pairs were common, supporting the inverse-composition hypothesis. However, local algebraic attacks did not recover the hidden RX pattern.

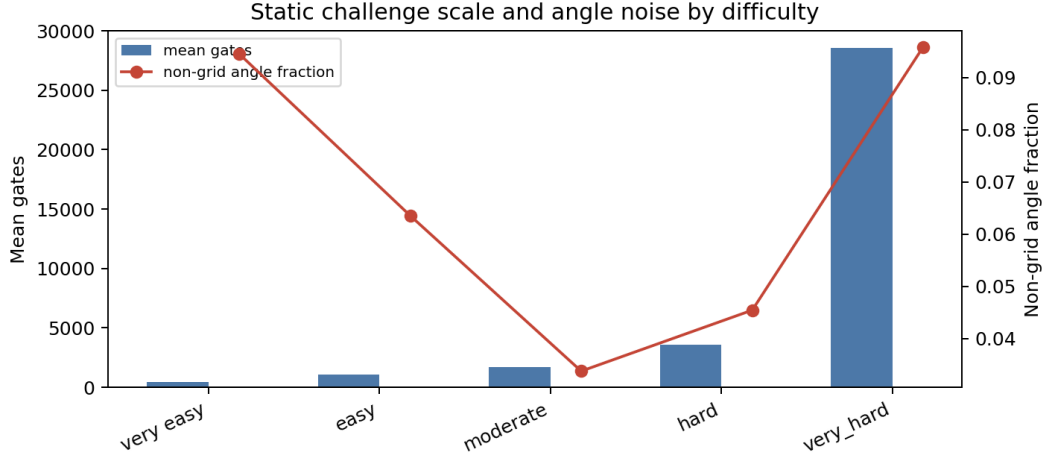


Figure 1: Static scale by difficulty. Very-hard circuits are much larger and use a distinct angle-noise profile, consistent with a stronger obfuscation pass.

Important negative findings:

- Native-basis Qiskit transpilation and adjacent cancellation barely reduced the circuits.
- Snapping angles to nearby π fractions helped syntax reduction slightly but did not reveal peak bits.
- $RX(\pi)$ remnants, SWAP tracking, and first/trailing RX -layer heuristics failed on small circuits where exact answers were known.
- The analysis supports nonlocal or globally dispersed pairings: the cancellations are not adjacent local patterns after obfuscation.

4 Exact Statevector Baseline

Exact Aer statevector simulation was run for circuits up to 28 qubits. This produced ten exact peaks and gave a ground truth for validating approximate methods. Statevector beyond this range was skipped because memory scales as $16 \cdot 2^n$ bytes before overhead.

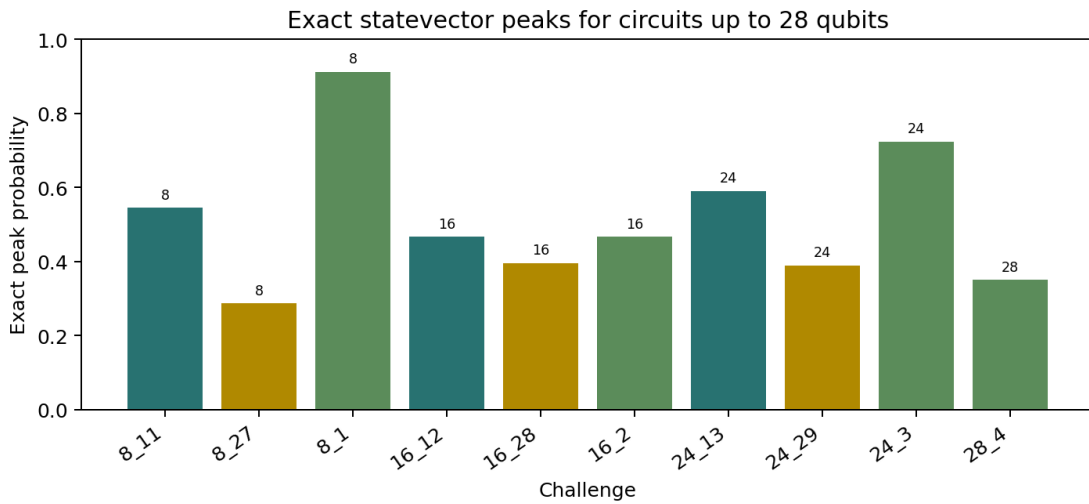


Figure 2: Exact peak probabilities for solved small circuits. The number above each bar is the qubit count.

Challenge	Difficulty	Qubits	Peak bitstring	Peak prob.	Gap
8_11	easy	8	01001110	0.545448	0.476764
8_27	moderate	8	11001001	0.287907	0.225213
8_1	very easy	8	10101101	0.912638	0.874726
16_12	easy	16	1111000101101011	0.466537	0.393572
16_28	moderate	16	1101001111011100	0.396206	0.373075
16_2	very easy	16	1010101011001000	0.466970	0.369383
24_13	easy	24	111110011111001011010001	0.591248	0.544080
24_29	moderate	24	110100010111100001001001	0.390097	0.376660
24_3	very easy	24	011110010000101010001000	0.723984	0.676891
28_4	very easy	28	1111111000101010110110011111	0.351112	0.277609

5 Aer MPS Sampling and Distillation

Aer MPS sampling was used in two ways. First, the requested IDs 11, 26, 34, 41, and 49 were run with statevector where feasible and MPS otherwise. Second, a pilot distillation grid ran six trials per selected circuit using combinations of `shots=512,2048,4096`, `bond.dim=16,32,64`, and two seeds.

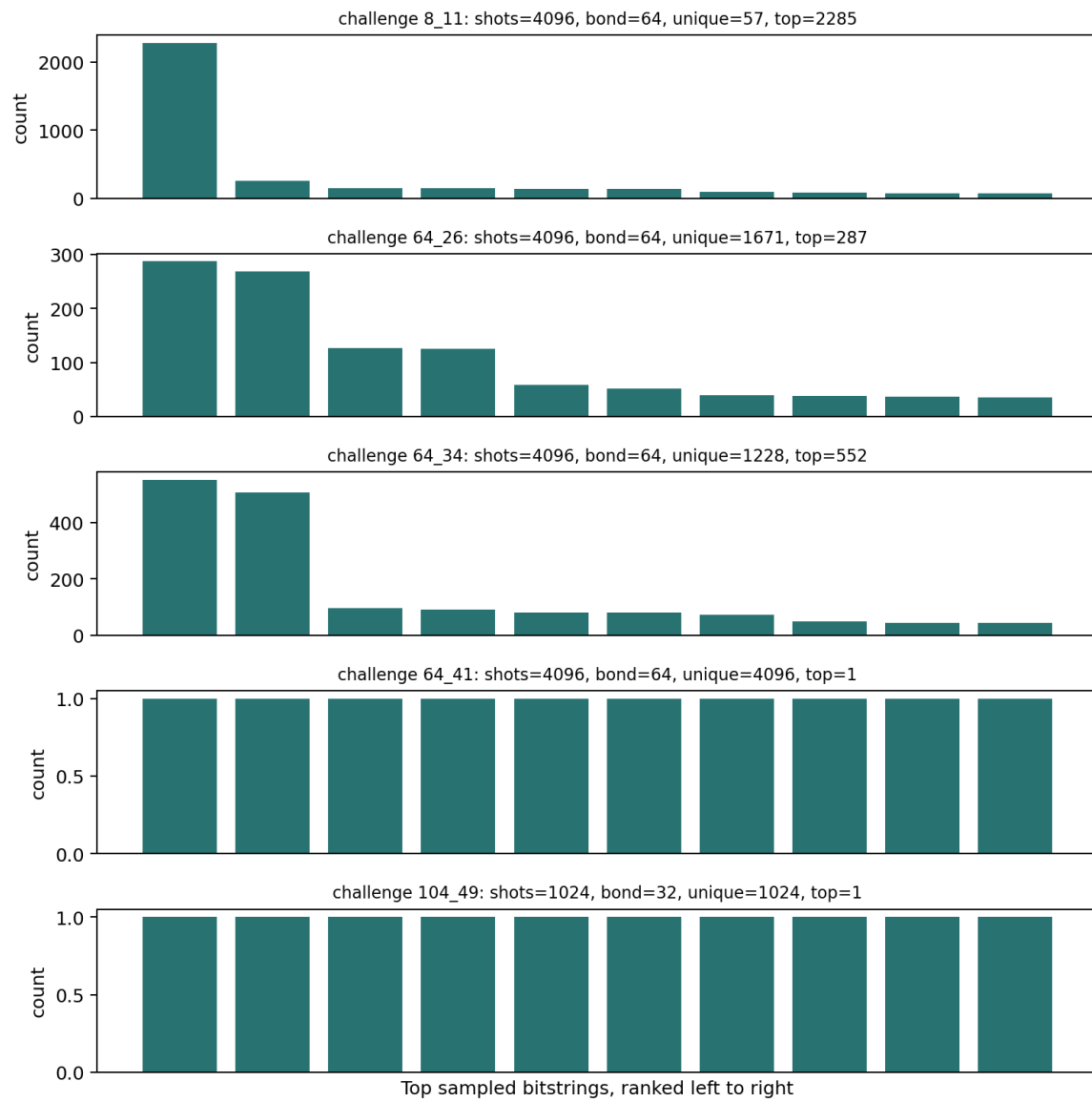


Figure 3: Top sampled counts for the requested MPS runs. IDs 26 and 34 have repeated top samples; IDs 41 and 49 are essentially all-unique samples and therefore low confidence.

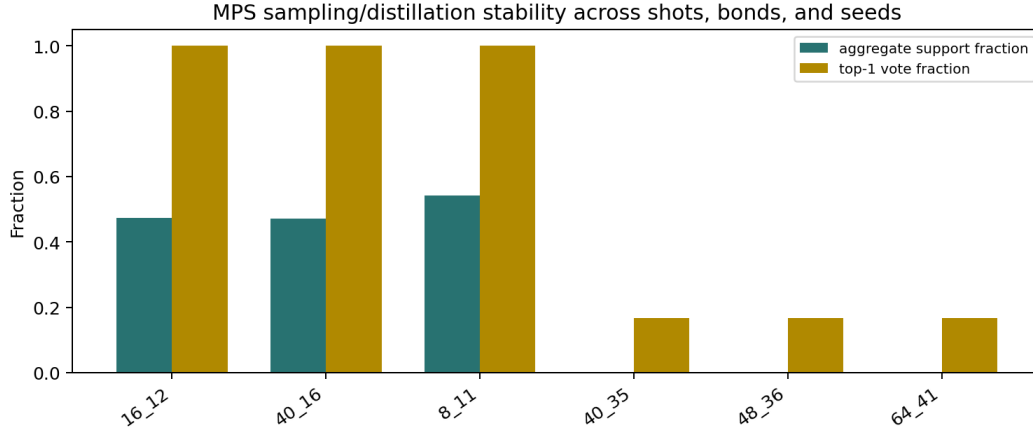


Figure 4: MPS distillation stability. Easy circuits and 40_16 were stable across seeds/settings; hard pilot circuits were unstable with flat aggregate support.

ID	Qubits	Shots	Bond	Top sampled bitstring	Count	Unique
8_11	8	4096	64	010011110	2285	57
64_26	64	4096	64	011010101010001101011101100001110001011011011001110 00110011001110	287	1671
64_34	64	4096	64	0011010100010011001110101110100101001011001011011100 11110111001110	552	1228
64_41	64	4096	64	001100110001000100010101000111011100100000000101110 0011011110010	1	4096
104_49	104	1024	32	11100011111101110111101011000110010011001001100101 000000011101000010101111011101000100000110101001101 01	1	1024

The MPS distillation summary was the most useful selection rule for non-exact circuits. For 40_16, all six trials selected the same candidate and the aggregate support fraction was about 0.471. For 40_35, 48_36, and 64_41, each top-1 candidate changed across runs and the aggregate winner had support of only two samples; those are not credible answers.

6 peaked-circuit-simulation Runs

The bundled `peaked-circuit-simulation` method was adapted through runtime compatibility shims for the installed Qiskit version. The runner used MPO compression plus unswapping, converted the compressed MPO to an MPS, and extracted a candidate by thresholding one-qubit marginals.

The main parameters used for the full sweep were:

```
max_bond=512, cutoff=0.002, unswap_threshold=1e6, max_its=10, rewire_trials=64, seed=123.
```

The full sweep was submitted as Slurm job 34607501 on `interruptible_gpu`; it produced 21 JSON files before cancellation: 12 completed `ok`, 9 remained `started`, and 28 challenge outputs were missing. A retry array 34611837 was also cancelled after about three minutes, so no additional retry outputs landed.

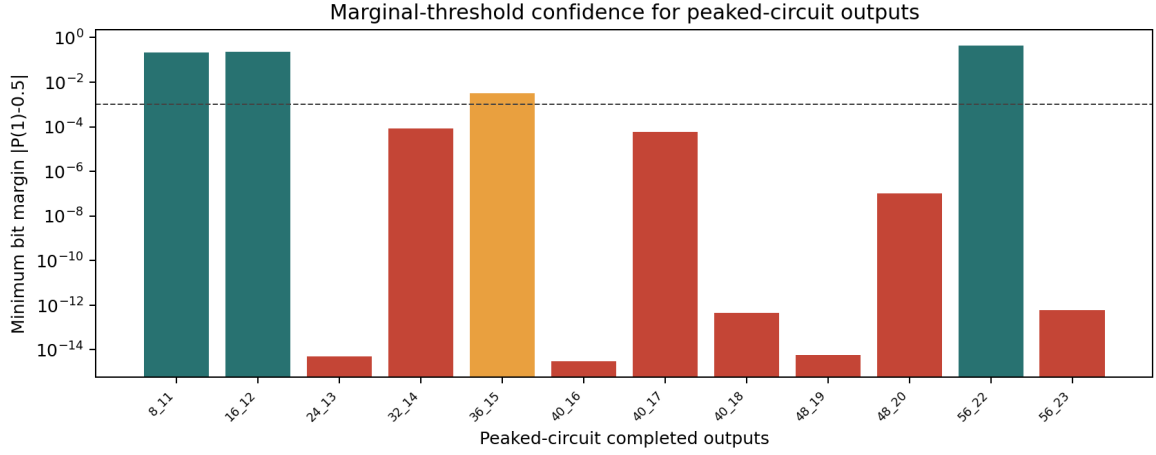
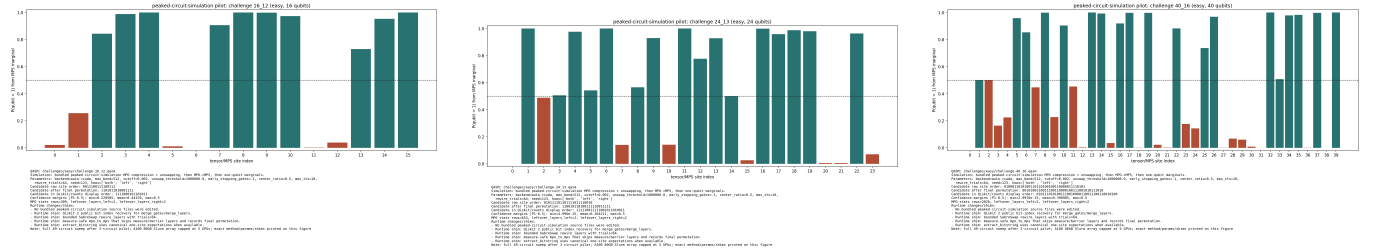


Figure 5: Confidence of completed **peaked-circuit-simulation** outputs. A small minimum margin means at least one inferred bit was near the $P(1) = 0.5$ decision boundary.



16_12: matched exact, healthy margin.

24_13: one-bit miss.

40_16: disagrees with stable MPS distill.

Figure 6: Representative peaked-circuit marginal plots. The images were generated by the runner and include method/parameter annotations.

Challenge	Difficulty	Qubits	Peaked candidate	Min margin	MPS bond
16_12	easy	16	1111000101101011	2.30e-01	4
24_13	easy	24	111110011111001011010011	5.00e-15	7
32_14	easy	32	0000011100110100010111011101100	8.25e-05	4
36_15	easy	36	101000011011111010101100111011001100	3.23e-03	6
40_16	easy	40	0101110101001110010000110011100110010100	3.00e-15	4
40_17	easy	40	0010010101010111001001000010001000001001	5.74e-05	4
40_18	easy	40	1110010011010011000001010001010101100011	4.63e-13	6
48_19	easy	48	011001010111101100111110000001011101001110010000	5.66e-15	3
48_20	easy	48	010111000101001011110000100000011000010000010010	1.01e-07	3
56_22	easy	56	111001001001100100111100101101100110000001000101111 11011	4.53e-01	2
56_23	easy	56	101000110000001010111010110111101111011100100000100 00001	6.00e-13	4
8_11	easy	8	01001110	2.10e-01	4

6.1 Why 24_13 Failed

The 24_13 exact statevector peak is

111110011111001011010001

with probability 0.591248. The peaked MPO/MPS run emitted

111110011111001011010011,

which differs in the second bit from the right, i.e. qubit q_1 .

This was not a genuine ambiguity in the exact circuit. Exact statevector gives $P(q_1 = 1) = 0.0265$, so the bit should clearly be 0. The compressed MPS/marginal extraction estimated the mapped marginal as $P(q_1 = 1) \approx 0.5068$, barely above the threshold, and therefore flipped the bit. The wrong candidate has exact probability about 0.01089 and rank roughly 12, so the failure was due to lossy compression/order-sensitive marginal extraction rather than a close second peak.

7 Cross-Method Validation and Answer Selection

The selection rule that emerged is:

1. Use exact statevector answers whenever available.
2. For non-exact circuits, prefer candidates repeated across seeds, shot counts, and bond dimensions.
3. Down-rank any one-qubit marginal answer with minimum margin near zero.
4. Treat all-unique MPS samples as samples only, not peak evidence.
5. For serious submission candidates, the next validation should estimate probabilities of a short candidate set directly rather than relying on one full distribution.

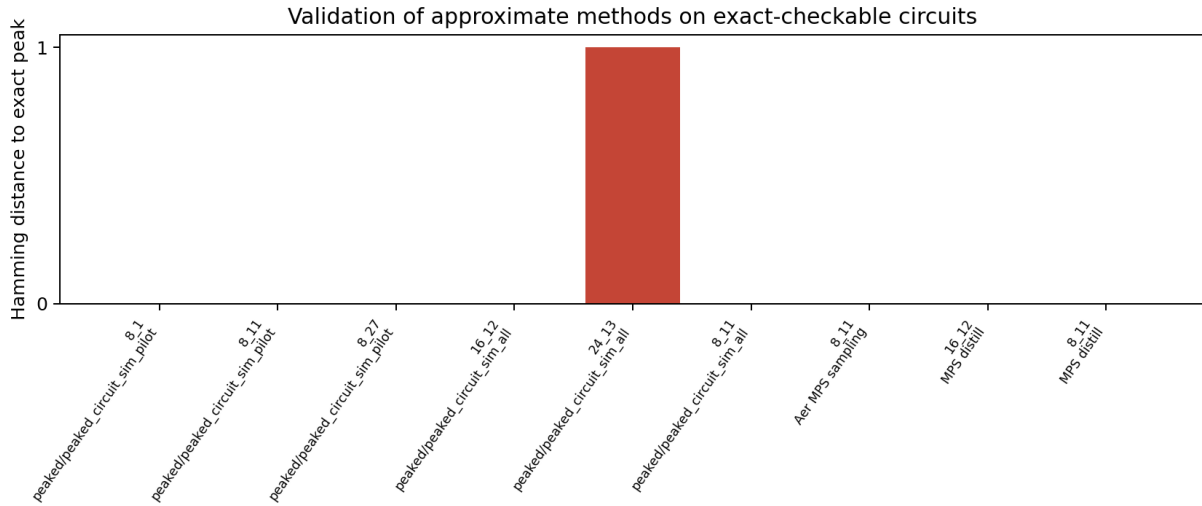


Figure 7: Approximate-method validation against exact baselines. The only confirmed mismatch was 24_13 from the peaked marginal method.

Table 1: Candidate bitstrings and confidence after this session.

Challenge	Source	Candidate	Confidence	Reason
8_1	exact/peaked	10101101	validated	Exact peak $p=0.9126$; peaked matched.
8_11	exact/peaked/MPS	01001110	validated	All methods matched; exact $p=0.5454$.
8_27	exact/peaked	11001001	validated	Exact $p=0.2879$; peaked matched.

Challenge	Source	Candidate	Confidence	Reason
16_12	exact/peaked/MPS	1111000101101011	validated	All methods matched; exact $p=0.4665$.
24_13	exact	111110011111001011010001	validated	Peaked marginal output was one bit off.
40_16	MPS distill	0101110101001110011000111011100110010110	strong candidate	Stable across 6/6 MPS trials; peaked disagreed by 3 bits.
64_26	Aer MPS	0110101010100011010111011000011100010110110110 011100011001100110	candidate	Top count 287/4096; runner candidate only.
64_34	Aer MPS	0011010100010011001110101110100101001011001011 011001111011100110	candidate	Top count 552/4096; runner candidate only.
64_41	MPS samples	–	unreliable	4096/4096 unique in selected run; distill unstable.
104_49	MPS samples	–	unreliable	1024/1024 unique; no repeated peak evidence.

8 Current Conclusions

The exact answers for small circuits are reliable. The `peaked-circuit-simulation` method is promising but the current marginal-threshold version is not submission-safe without validation: it matched several exact circuits but missed 24_13, and it disagreed with a stable MPS-distilled 40_16 candidate. For larger problems, the best current candidates are 40_16, 64_26, and 64_34; 64_41 and 104_49 remain low confidence because sampled counts were flat.

Recommended next work:

- Resubmit the incomplete peaked sweep only after the Slurm cancellation issue is understood.
- Add candidate probability evaluation: contract or sample around a small candidate set instead of extracting only one-qubit marginals.
- Continue higher-shot/higher-bond MPS distillation for hard circuits, requiring repeated winners across seeds.
- Pursue tree/tensor-network decompositions separately for nonlocal pairings and very-hard dense circuits.

References

- [1] H. Gharibyan et al., *Heuristic Quantum Advantage with Peaked Circuits*, arXiv:2510.25838, 2025.
- [2] D. Kremer and E. Dupuis, *Efficient Classical Simulation of Heuristic Peaked Quantum Circuits*, arXiv:2604.21908, 2025.
- [3] M. Rudolph and J. Tindall, *Simulating and Sampling from Quantum Circuits with 2D Tensor Networks*, arXiv:2507.11424, 2025.